

EXAM 7 – SPRING 2014

22. (3.25 points)

An insurer has decided to calculate risk-based performance metrics for its two lines of business. It has chosen a risk capital metric of 99th percentile VaR. Below is a summary of results (\$000,000) for the two lines of business.

<u>Line of Business</u>	<u>Expected Annualized Profit</u>	<u>Coefficient of Variation</u>
A	50	300%
B	9	250%

- $\Phi^{-1}(0.99) = 2.326$
- Income follows a normal distribution.
- The lines of business are independent.

a. (0.75 point)

Determine which line of business performs better using return on risk-adjusted capital.

b. (0.75 point)

Calculate the return on risk capital and the risk-adjusted performance metric for the portfolio as a whole.

c. (0.5 point)

Allocate the risk capital to both portfolios using the proportional allocation method.

d. (1.25 points)

Calculate the allocation of capital using an incremental marginal approach and discuss an advantage this approach has over the proportional allocation method.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 22	
TOTAL POINT VALUE: 3.25	LEARNING OBJECTIVE: C4: Demonstrate the properties of various risk measures and their limitations
SAMPLE ANSWERS	
Part a: 0.75 point	
<i>Sample Answer 1</i> Standard dev A = $3 \times 50 = 150$; B = $2.5 \times 9 = 22.5$ VaRA = $50 + 2.326 \times 150 = 398.9$ $50/398.9 = 12.5\%$ B = VaR = $9 + 2.326 \times 22.5 = 61.34$ $9/61.34 = 14.7\%$; B has a higher return on risk adj capital; B better <i>Sample Answer 2</i> Use that sigma = cov $\times$ mean, then use that capital required = $\sigma \times z$ $\sigma_A = 50 \times 3 = 150$; $\sigma_B = 9 \times 2.5 = 22.5$ capital_A = $150 \times 2.326 = 348.9$; capital_B = $22.5 \times 2.326 = 52.34$ return_A = $50/348.9 = 14.3\%$, return_B = $9/52.34 = 17.2\%$; return B better <i>Sample Answer 3</i> Line B performs better because it has a lower CV	
Part b: 0.75 point	
<i>Sample Answer 1</i> Standard dev = $(150^2 + 22.5^2) = 151.68$ VaR = $50 + 9 + 2.326 \times 151.68 = 411.81$ $59/411.81 = 14.3\%$ <i>Sample Answer 2</i> $\sigma_{\text{Total}} = (150^2 + 22.5^2) = 151.68$ capital = $151.2 \times 2.326 = 352.8$ return = $59/352.8 = 16.7\%$	
Part c: 0.5 point	
<i>Sample Answer 1</i> Allocation to A = $411.81M \times (398.9/(398.9+61.335)) = 357M$ Allocation to B = $411.81M \times (61.335/(398.9+61.335)) = 55M$	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 2

Allocation to A = $352.8 \times (348.9 / (348.9 + 52.34)) = 306$

Allocation to B = $352.8 \times (52.34 / (348.9 + 52.34)) = 46$

Part d: 1.25 points

Sample Answer 1

Last in Marginal Allocation A = $411.81 - 61.335 = 350$

Allocate to A = $411.81 \times (350.475 / 363.385) = 397M$

B: $411.8 - 398.9 = 15$

Allocate to B = $411.81 \times (12.91 / 363.385) = 14.63M$

This method measures the marginal impact of the addition of each business unit treating it as the last one in. The method has the advantage of maintaining the financial principle of comparing marginal profits with marginal costs.

Sample Answer 2

RORAC = Expected Profit / Portfolio VaR where

	Portfolio VaR	LOB VaR	Inc. Marg. VaR	% Alloc	Marg. VaR
LOB A	352.8	348.9	300.5	99%	348.3
LOB B	352.8	52.3	3.9	1%	4.5
Total			304.4		352.8

Advantages of incremental marginal approach over the proportional allocation method:

- Maintains financial principle of comparing marginal profits with marginal costs
- Marginal allocation charges each unit with the additional capital it requires
- Suitable allocation, which means increasing business with highest ROE will increase ROE for entire company

EXAMINER'S REPORT

Based on the learning objective, the candidate needed to demonstrate understanding of the properties of VaR. Part d was more advanced, which was evident in that the majority of candidates did not receive full credit for this part and many did not attempt the calculation. The information that the candidate needed to successfully answer the question was based on multiple papers on the syllabus and required the candidate to draw from the various papers.

For solutions 1 and 2 above, it is necessary to calculate the allocated capital in order to compare the risk adjusted return. Two different calculations of allocated capital were accepted, resulting in the two different solutions. For solution 1 above the allocated capital was calculated as expected

SAMPLE ANSWERS AND EXAMINER'S REPORT

profit plus 2.326 standard deviations. For solution 2 the allocated capital was calculated as 2.326 standard deviations.
Part a
Over half of the candidates got full credit for this part. A number of candidates incorrectly calculated allocated capital resulting in a response with negative allocated capital.
Part b
A common error in part b was candidates calculating the portfolio allocated capital as a sum of the allocated capital for Lines A & B.
Part c
A common error in part c was candidates not using the allocated capital by line to allocate the portfolio risk capital.
Part d
Many candidates did not attempt the calculation. Very few candidates got full credit for this part.